Implementation Note: Non-Obvious Design Choices

Speech Assignment 2

GitHub: avantimittal/speech-assignment-2-hinglish-to-maithili-translation-

I. Q1: Hinglish ASR + Language Identification

Choice: Weighted Dual-Head Loss (0.7 / 0.3) over Equal Weighting

The obvious approach is a 0.5/0.5 split between frame-level ($\mathcal{L}_{\text{frame}}$) and segment-level ($\mathcal{L}_{\text{seg}}$) cross-entropy. This is **incorrect** for code-switching detection.

The combined loss is:

$$\mathcal{L} = 0.7\,\mathcal{L}_{\text{frame}} + 0.3\,\mathcal{L}_{\text{seg}}$$

The segment-level loss has *lower variance* (one label per $T = 200$-frame sequence vs. 200 per-frame labels), causing it to dominate gradient updates in later epochs under equal weighting. This starves the frame head of signal precisely when boundary-precision matters most.

Ablation result: Equal weighting (0.5/0.5) yields macro F1 = 0.79; the 0.7/0.3 split yields **F1 = 0.9217**, a 13 pp gain — a non-trivial improvement from a single hyperparameter.

Choice: Flat β Bias over Context-Sensitive N-gram Bias

The bigram LM computes $P_{\text{SB}}(w \mid w_{t-1})$ at the *word* level. Whisper decodes at the *BPE subword* level (51,865-token vocabulary). Mapping a partially completed BPE subword sequence to a word-level bigram context requires non-trivial aggregation over partial completions that is not implemented in the HuggingFace `LogitsProcessor` API.

Decision: Deploy a flat additive bias $\ell'_t(x) = \ell_t(x) + \beta$ for $x \in \mathcal{T}$, with $\beta = 1.5$ (boosting probability by $e^{1.5} \approx 4.48\times$). This gives a principled domain-prior without requiring BPE-to-word context alignment. Applied to 117 technical token IDs. Bigram sentence perplexity on in-domain test: **9.95**.

II. Q2: Phonetic Mapping & Translation to Maithili

Choice: Sentence-Wise Translation with Longest-First Pre-substitution

The direct approach — passing the entire 8,234-character Hinglish transcript to NLLB-200 in one call — fails in practice for three reasons: (1) it exceeds the 256 subword token limit causing silent truncation, (2) sequence-to-sequence models degrade on very long inputs in low-resource language pairs, and (3) any GPU OOM failure loses all progress.

Decision:

- Split on Devanagari danda, full stop, and punctuation → 79 sentences.
- Apply 82-term technical dictionary via regex *longest-match first* before neural decoding: $\hat{s} = \text{RegexSub}(s,\ \mathcal{D}_{\text{tech}})$.
- Longest-first ordering is non-obvious: without it, "science" is substituted inside "neuroscience", corrupting the longer term.
- Translate each sentence independently, concatenate outputs.

This makes the pipeline **fault-tolerant**: a single failed sentence (1 of 79 skipped) does not abort the full transcript.

III. Q3: Zero-Shot Voice Cloning + Prosody Transfer

Choice: Sample-Level DTW Warp over Phase-Vocoder Time Stretching

After XTTS-v2 synthesis, two prosody-transfer approaches are possible: (a) **Phase vocoder** time-stretching — smooth but introduces spectral smearing at consonant onsets that degrades Hindi stop consonants; (b) **Sample-level DTW index remapping** — sharper consonants but may introduce sample-level discontinuities.

Decision: Use FastDTW to align reference and synthesised F0 contours, then remap synthesised audio at sample level: $n_k = j_k \cdot H_{\text{pyin}}$ ($H_{\text{pyin}} = 512$), followed by a **median filter (kernel 5)** to smooth discontinuities.

Result: MCD drops from 9.4 dB (flat XTTS) to **6.18 dB** (DTW-warped), F0 RMSE from 38.2 Hz to 21.7 Hz. Consonant sharpness is preserved — critical for Maithili retroflex contrasts absent in Hindi.

Choice: Runtime Monkey-Patch for `transformers` Compatibility

XTTS-v2 imports `isin_mps_friendly` from `transformers.pytorch_utils`, removed in `transformers`≥4.41. Downgrading `transformers` would break NLLB-200. **Decision:** Inject a wrapper around `torch.isin` at runtime before any TTS import — keeping the entire pipeline on a single `transformers` version.

IV. Q4: Anti-Spoofing + Adversarial Robustness

Choice: LFCC over MFCC for Spoof Detection

MFCC uses mel-scale filterbanks that compress high-frequency energy above 4 kHz — exactly where neural vocoders (including XTTS-v2) leave characteristic synthesis artifacts. Using MFCC for spoof detection therefore discards the most discriminative signal.

Decision: Use 40-dimensional **LFCC** (linear filterbank, $N_{\text{FFT}} = 512$, $H = 160$), which preserves full high-frequency detail. MLP classifier trained on 30 real + 30 spoof 1-second chunks for 40 epochs. **Achieved EER: 2.45%** vs. requirement of <10% — a 4× safety margin on limited training data.

Choice: Minimum-ε Grid Search over Fixed Attack

Rather than fixing ε a priori (the standard FGSM setup), the implementation searches for the *minimum* ε satisfying *both* SNR > 40 dB and prediction flip > 0.1:

$$\varepsilon^* = \min_{\varepsilon} \text{ s.t. } \mathrm{SNR}(\varepsilon) > 40\,\mathrm{dB} \;\wedge\; |\hat{y}_{\mathrm{adv}} - \hat{y}_{\mathrm{orig}}| > 0.1$$

This reveals the model's *exact* vulnerability threshold rather than a conservative upper bound. Result: $\varepsilon^* = 1.2{\times}10^{-4}$, **SNR = 78.21 dB** — 38 dB above the 40 dB requirement, demonstrating that the BiLSTM LID is vulnerable to imperceptible perturbations far below human perceptual thresholds.

V. Q5: Evaluation Metrics

Choice: Multi-Metric Evaluation over Single WER

Using only WER would reward ASR accuracy while ignoring synthesis quality, speaker similarity, and security robustness. A system with low WER but high MCD (unnatural voice) or high EER (trivially fooled by cloned audio) would still pass a WER-only gate.

Decision: Assign a distinct metric to each sub-task: WER (ASR), Macro F1 (LID boundary accuracy), MCD (voice similarity), EER (spoof detection), and SNR (adversarial imperceptibility). This provides *orthogonal* coverage: no single metric can mask failure in another subsystem. All five requirements are met as summarised below.

Metric	Required	Achieved
LID Macro F1	≥0.85	**0.9217**
English WER	<15%	**13.20%**
Hindi WER	<25%	**14.80%**
MCD	<8.0 dB	**6.18 dB**
EER	<10%	**2.45%**
FGSM SNR	>40 dB	**78.21 dB**

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